1 Code in FeynCalc

Listing 1: FeynCalc code

```
ClearAll["Global`*"];
Needs["FeynCalc`"]
(*define denomenators*)
denS[s_] := s - M_A^2;
denT[t_] := t - M_A^2;
denU[u_] := u - M_A^2;
SetM_Andelstam[s, t, u, p1, p2, p3, p4, mchi, mchi, mchi
, mchi];
(* Process 1:\[Chi]     \[RightArrow]\[Chi]      *)
M1s = gchi^2/denS[s]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorU[p1, mchi
]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
];
M1t = gchi^2/denT[t]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p1, mchi
]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
];
M1 = M1s - M1t;
M1sq = (FermionSpinSum[M1*ComplexConjugate[M1]])/4 //
 Contract //
    DiracSimplify // Simplify;
(* Process 2:\[Chi] \[Chi]\[RightArrow]\[Chi] \[Chi]
*)
M2t = gchi^2/denT[t]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p1, mchi
]*
   SpinorUBar[p4, mchi] . GA[\[Mu]] . SpinorU[p2, mchi
];
M2u = gchi^2/denU[u]*
   SpinorUBar[p4, mchi] . GA[\[Mu]] . SpinorU[p1, mchi
]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p2, mchi
];
M2 = M2t - M2u;
M2sq = (FermionSpinSum[M2*ComplexConjugate[M2]])/4 //
 Contract //
    DiracSimplify // Simplify;
(* Process 3:          \[RightArrow]          *)
M3t = gchi^2/denT[t]*
   SpinorVBar[p1, mchi] . GA[\[Mu]] . SpinorV[p3, mchi
]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
];
M3u = gchi^2/denU[u]*
   SpinorVBar[p1, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorV[p3, mchi
];
M3 = M3t - M3u;
M3sq = (FermionSpinSum[M3*ComplexConjugate[M3]])/4 //
 Contract //
    DiracSimplify // Simplify;
ClearAll[ "Global`*"];
Needs["FeynCalc`"]
denS[s_] := s - MA^2;
denT[t_] := t - MA^2;
```

```
denU[u_] := u - MA^2;
(* Process:\[Mu]-\[Mu]+ ->\[Chi] \[Chi]bar*)
FCClearScalarProducts[];
SetMandelstam[s, t, u, p1, p2, p3, p4, mmu, mmu, mchi
, mchi];
M4 = (gmu*gchi)/denS[s]*
   SpinorVBar[p2, mmu] . GA[\[Mu]] . SpinorU[p1, mmu]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
];
M4sqClean = (FermionSpinSum[M4*ComplexConjugate[M4]])
/4 // Contract //
     DiracSimplify // Simplify;
(* Process 5:\[Tau]-\[Tau]+ ->\[Chi] \[Chi]bar*)
FCClearScalarProducts[];
SetMandelstam[s, t, u, p1, p2, p3, p4, mtau, mtau, mchi
, mchi];
M5 = (gtau*gchi)/denS[s]*
   SpinorVBar[p2, mtau] . GA[\[Mu]] . SpinorU[p1, mtau
]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
];
M5sqClean = (FermionSpinSum[M5*ComplexConjugate[M5]])
/4 // Contract //
     DiracSimplify // Simplify;
(* Process 6:\[Nu]_\[Mu] \[Nu]bar_\[Mu]->\[Chi] \[Chi
]bar*)
FCClearScalarProducts[];
SetMandelstam[s, t, u, p1, p2, p3, p4, 0, 0, mchi, mchi
];
M6 = (gnumu*gchi)/denS[s]*
   SpinorVBar[p2, 0] . GA[\[Mu]] . (1 - GA[5]) . SpinorU
[p1, 0]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
];
M6sqClean = (FermionSpinSum[M6*ComplexConjugate[M6]])
/4 // Contract //
     DiracSimplify // Simplify;
(* Process 7:\[Nu]_\[Tau] \[Nu]bar_\[Tau]->\[Chi] \[Chi
]bar*)
FCClearScalarProducts[];
SetMandelstam[s, t, u, p1, p2, p3, p4, 0, 0, mchi, mchi
];
M7 = (gnutau*gchi)/denS[s]*
   SpinorVBar[p2, 0] . GA[\[Mu]] . (1 - GA[5]) . SpinorU
[p1, 0]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi
];
M7sqClean = (FermionSpinSum[M7*ComplexConjugate[M7]])
/4 // Contract //
     DiracSimplify // Simplify;
Print["Process1 (\[Chi] \[Chi]bar -> \[Chi] \[Chi]bar
): ",
  M1sq // Apart[#, {s, t}] &];
Print["Process2 (\[Chi] \[Chi] -> \[Chi] \[Chi]): ",
  M2sq // Apart[#, {t, u}] &];
Print["Process3 (\[Chi]bar \[Chi]bar -> \[Chi]bar \[Chi
]bar): ",
  M3sq // Apart[#, {t, u}] &];
Print["Process4 (\[Mu]- \[Mu]+ -> \[Chi] \[Chi]bar):
", M4sq];
Print["Process5 (\[Tau]- \[Tau]+ -> \[Chi] \[Chi]bar)
: ", M5sq];
```

```
Print["Process6 (v_\[Mu] v_\[Mu]bar -> \[Chi] \[Chi]bar
): ", M6sq];
Print["Process7 (v_\[Tau] v_\[Tau]bar -> \[Chi] \[Chi
]bar): ", M7sq];
```

2 List of Processes

For only the leading order. Taking χ as Dirac Fermion, not considering real scaler or Majorana. Mediator is vector A', not scaler.

1. $\chi\,\bar{\chi} \to \chi\,\bar{\chi}$ (s- and t-channel)
2. $\chi\,\chi \to \chi\,\chi$ (t- and u-channel)
3. $\bar{\chi}\,\bar{\chi} \to \bar{\chi}\,\bar{\chi}$ (t- and u-channel)
4. $\mu^-\mu^+ \to \chi\,\bar{\chi}$ (only s-channel)
5. $\tau^-\tau^+ \to \chi\,\bar{\chi}$ (only s-channel)
6. $\nu_\mu\,\bar{\nu}_\mu \to \chi\,\bar{\chi}$ (only s-channel, left-handed neutrino)
7. $\nu_\tau\,\bar{\nu}_\tau \to \chi\,\bar{\chi}$ (only s-channel, left-handed neutrino)

3 Outputs

$$
\begin{aligned}
&\text{Process 1 } (\chi\bar{\chi} \to \chi\bar{\chi}):\ \Bigg\{ -\frac{2\left(-24m_\chi^4 + 4M_A^2 m_\chi^2 + 4um_\chi^2 - u^2\right) g_\chi^4}{\left(M_A^2 - t\right)^2} \\
&+ \frac{2s^2 g_\chi^4}{\left(M_A^2 - t\right)^2} - \frac{8m_\chi^2 s g_\chi^4}{\left(M_A^2 - t\right)^2} \\
&+ \frac{2\left(8m_\chi^4 g_\chi^4 + 4M_A^2 m_\chi^2 g_\chi^4 + t^2 g_\chi^4 + u^2 g_\chi^4 - 4m_\chi^2 t g_\chi^4 - 4m_\chi^2 u g_\chi^4\right)}{\left(s - M_A^2\right)^2} \\
&- \frac{4\left(4m_\chi^4 g_\chi^4 + u^2 g_\chi^4 - 2m_\chi^2 u g_\chi^4\right)}{\left(s - M_A^2\right)\left(M_A^2 - t\right)}, \frac{2t^2 g_\chi^4}{\left(M_A^2 - s\right)^2} \\
&- \frac{8m_\chi^2 t g_\chi^4}{\left(M_A^2 - s\right)^2} + \frac{2\left(8m_\chi^4 + 4M_A^2 m_\chi^2 - 4um_\chi^2 + u^2\right) g_\chi^4}{\left(M_A^2 - s\right)^2} \\
&- \frac{2\left(-24m_\chi^4 g_\chi^4 + 4M_A^2 m_\chi^2 g_\chi^4 - s^2 g_\chi^4 - u^2 g_\chi^4 + 4m_\chi^2 s g_\chi^4 + 4m_\chi^2 u g_\chi^4\right)}{\left(t - M_A^2\right)^2} \\
&- \frac{4\left(4m_\chi^4 g_\chi^4 + u^2 g_\chi^4 - 2m_\chi^2 u g_\chi^4\right)}{\left(M_A^2 - s\right)\left(t - M_A^2\right)} \Bigg\}
\end{aligned}
\tag{1}
$$

Process 2 ($\chi\chi \to \chi\chi$): $\left\{ \frac{2\left(24m_\chi^4 + 4M_A^2 m_\chi^2 - 4sm_\chi^2 - 8um_\chi^2 + s^2\right) g_\chi^4}{\left(M_A^2 - u\right)^2} \right.$ (2)

$+ \frac{2t^2 g_\chi^4}{\left(M_A^2 - u\right)^2} - \frac{8m_\chi^2 t g_\chi^4}{\left(M_A^2 - u\right)^2} - \frac{4\left(20m_\chi^4 g_\chi^4 + s^2 g_\chi^4 - 6m_\chi^2 s g_\chi^4 - 4m_\chi^2 u g_\chi^4\right)}{\left(t - M_A^2\right)\left(M_A^2 - u\right)}$

$- \frac{2\left(-24m_\chi^4 g_\chi^4 + 4M_A^2 m_\chi^2 g_\chi^4 - s^2 g_\chi^4 - u^2 g_\chi^4 + 4m_\chi^2 s g_\chi^4 + 4m_\chi^2 u g_\chi^4\right)}{\left(t - M_A^2\right)^2}, \frac{2u^2 g_\chi^4}{\left(M_A^2 - t\right)^2}$

$+ \frac{2\left(24m_\chi^4 + 4M_A^2 m_\chi^2 - 4sm_\chi^2 - 8tm_\chi^2 + s^2\right) g_\chi^4}{\left(M_A^2 - t\right)^2}$

$- \frac{8m_\chi^2 u g_\chi^4}{\left(M_A^2 - t\right)^2} - \frac{4\left(20m_\chi^4 g_\chi^4 + s^2 g_\chi^4 - 6m_\chi^2 s g_\chi^4 - 4m_\chi^2 t g_\chi^4\right)}{\left(M_A^2 - t\right)\left(u - M_A^2\right)}$

$\left. - \frac{2\left(-24m_\chi^4 g_\chi^4 + 4M_A^2 m_\chi^2 g_\chi^4 - s^2 g_\chi^4 - t^2 g_\chi^4 + 4m_\chi^2 s g_\chi^4 + 4m_\chi^2 t g_\chi^4\right)}{\left(u - M_A^2\right)^2} \right\}$

Process 3 ($\bar{\chi}\bar{\chi} \to \bar{\chi}\bar{\chi}$): $\left\{ \frac{2\left(24m_\chi^4 + 4M_A^2 m_\chi^2 - 4sm_\chi^2 - 8um_\chi^2 + s^2\right) g_\chi^4}{\left(M_A^2 - u\right)^2} \right.$ (3)

$+ \frac{2t^2 g_\chi^4}{\left(M_A^2 - u\right)^2} - \frac{8m_\chi^2 t g_\chi^4}{\left(M_A^2 - u\right)^2} - \frac{4\left(20m_\chi^4 g_\chi^4 + s^2 g_\chi^4 - 6m_\chi^2 s g_\chi^4 - 4m_\chi^2 u g_\chi^4\right)}{\left(t - M_A^2\right)\left(M_A^2 - u\right)}$

$- \frac{2\left(-24m_\chi^4 g_\chi^4 + 4M_A^2 m_\chi^2 g_\chi^4 - s^2 g_\chi^4 - u^2 g_\chi^4 + 4m_\chi^2 s g_\chi^4 + 4m_\chi^2 u g_\chi^4\right)}{\left(t - M_A^2\right)^2}, \frac{2u^2 g_\chi^4}{\left(M_A^2 - t\right)^2}$

$+ \frac{2\left(24m_\chi^4 + 4M_A^2 m_\chi^2 - 4sm_\chi^2 - 8tm_\chi^2 + s^2\right) g_\chi^4}{\left(M_A^2 - t\right)^2}$

$- \frac{8m_\chi^2 u g_\chi^4}{\left(M_A^2 - t\right)^2} - \frac{4\left(20m_\chi^4 g_\chi^4 + s^2 g_\chi^4 - 6m_\chi^2 s g_\chi^4 - 4m_\chi^2 t g_\chi^4\right)}{\left(M_A^2 - t\right)\left(u - M_A^2\right)}$

$\left. - \frac{2\left(-24m_\chi^4 g_\chi^4 + 4M_A^2 m_\chi^2 g_\chi^4 - s^2 g_\chi^4 - t^2 g_\chi^4 + 4m_\chi^2 s g_\chi^4 + 4m_\chi^2 t g_\chi^4\right)}{\left(u - M_A^2\right)^2} \right\}$

Process 4 ($\mu^-\mu^+ \to \chi\bar{\chi}$): $\frac{2g_\chi^2 g_\mu^2\left(2m_\chi^4 + 2m_\chi^2\left(2m_\mu^2 + s - t - u\right) + 2m_\mu^4 + 2m_\mu^2(s - t - u) + t^2 + u^2\right)}{\left(M_A^2 - s\right)^2}$ (4)

Process 5 ($\tau^-\tau^+ \to \chi\bar{\chi}$): $\frac{2g_\chi^2 g_\tau^2\left(2m_\chi^4 + 2m_\chi^2\left(2m_\tau^2 + s - t - u\right) + 2m_\tau^4 + 2m_\tau^2(s - t - u) + t^2 + u^2\right)}{\left(M_A^2 - s\right)^2}$ (5)

Process 6 ($\nu_\mu\bar{\nu}_\mu \to \chi\bar{\chi}$): $\frac{\left(4g_\chi^2 g_{\nu_\mu}^2\left(2m_\chi^4 + 2m_\chi^2(s - t - u) + t^2 + u^2\right)\right)}{\left(M_A^2 - s\right)^2}$ (6)

$$\text{Process 7 } (\nu_\tau \bar{\nu}_\tau \to \chi\bar{\chi}): \quad \frac{4g_\chi^2 g_{\nu_\tau}^2 \left(2m_\chi^4 + 2m_\chi^2(s-t-u) + t^2 + u^2\right)}{\left(M_A^2 - s\right)^2} \tag{7}$$

In process 1,2,3, things before comma and after comma are the same but different types of simplification.

4 Checks

1. Dimension: $[m^2]$=[s,t,u], $[|M|^2]$=$\frac{[m^4]}{[m^4]}$=1

2. Process 2 and 3 are the same results. Charge conjugation

3. Process 1 under substitution $\chi \to e$ is Bhabha scattering. Let $m_\chi \to m_e \to 0, g_\chi \to g_e = e, M_A \to M_\gamma = 0$, collapse to $2e^4(\frac{(u^2+s^2)}{t^2} + \frac{t^2+u^2}{s^2} + \frac{2u^2}{st})$ which is the same as the result of Bhabha scattering.

   Previously my result for Process 1 is $2e^4(\frac{(u^2+s^2)}{t^2} + \frac{t^2+u^2}{s^2} - \frac{2u^2}{st})$*, I believe it was wrong. I think a simple sign difference may be determined according to a simple sign problem. It is just* $s \to -s$ *or* $t \to -t$*. In the previous code of process 1, my M1 is M1s + M1t, which I now think is M1s - M1t, because of Wick's Theorem. After this the result is correct. Please check if this is correct.*

4. For Process 4,5,6,7, they are in the same form. But when I tried to substitute μ to ν_μ for Process 4, I got $\frac{1}{2}$ of Process 6. I realized that this is because neutrinos are only left-handed, only one possibility, while fermions can have an average coefficient of $\frac{1}{4}$. However the traces of GA and $\frac{(1-GA[5]}{2}$ have a ratio 2. So *4/2=2.

5. In my code, I used (1 - GA[5]) without an $\frac{1}{2}$ coefficient and still using $\frac{1}{4}$ coefficient as average coefficient. So the errors cancel out, leaving with correct values.

6. For process 5, substitute $\chi \to \mu, \tau \to e, A' \to \gamma$, while $m_e \to 0, M_\gamma = 0$, get $\frac{2e^4[2m_\mu^4+2m_\mu^2(s-t-u)+t^2+u^2]}{s^2}$. The standard result for $e\bar{e} \to \mu\bar{\mu}$ is $\frac{2e^4[(t-m_\mu^2)^2+(u-m_\mu^2)^2+2m_\mu^2 s]}{s^2} = \frac{2e^4(t^2-2m_\mu^2 t+m_\mu^4+u^2-2m_\mu^2 u+m_\mu^4+2m_\mu^2 s)}{s^2}$ $= \frac{2e^4(t^2+u^2+2m_\mu^2(s-t-u)+2m_\mu^4)}{s^2}$. So it is correct. And then can assert 4,6,7 are correct based on point 4 above.

7. In papers, first it is hard to find some papers that includes the results. I guess authors tend to have σs instead of matrix elements. I also searched Peskin, there is no

$$|M|^2(s,t,u)$$

but $|M|^2(p_1, p_2, p_3, p_4)$ there. They use different notations for p_is. Based on this, *I think I need some guidance on finding this in papers.*

8. Please give me other ways to check the result is correct.

9. I guess next should be calculating σ_T and $< \sigma v > (T)$. Or do you need more calculations for

$$|M|^2$$

like for Majorana fermions or other mediators? And I also want to know is it in general the above 7 results are not able to be found on other sources? Because I think they are very general.

Based on above, when do you think is a good time to meet?